

Cloud and Network Security.

Monday, July 15th, 2025.

**Network Security Groups and Application Security Groups
(Microsoft Azure).**

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Introduction

In this lab, I implement a virtual networking infrastructure in Azure to support two groups of servers: **Web Servers** and **Management Servers**. Each group is placed in its own **Application Security Group (ASG)** to enable role-based traffic control.

The configuration ensures that:

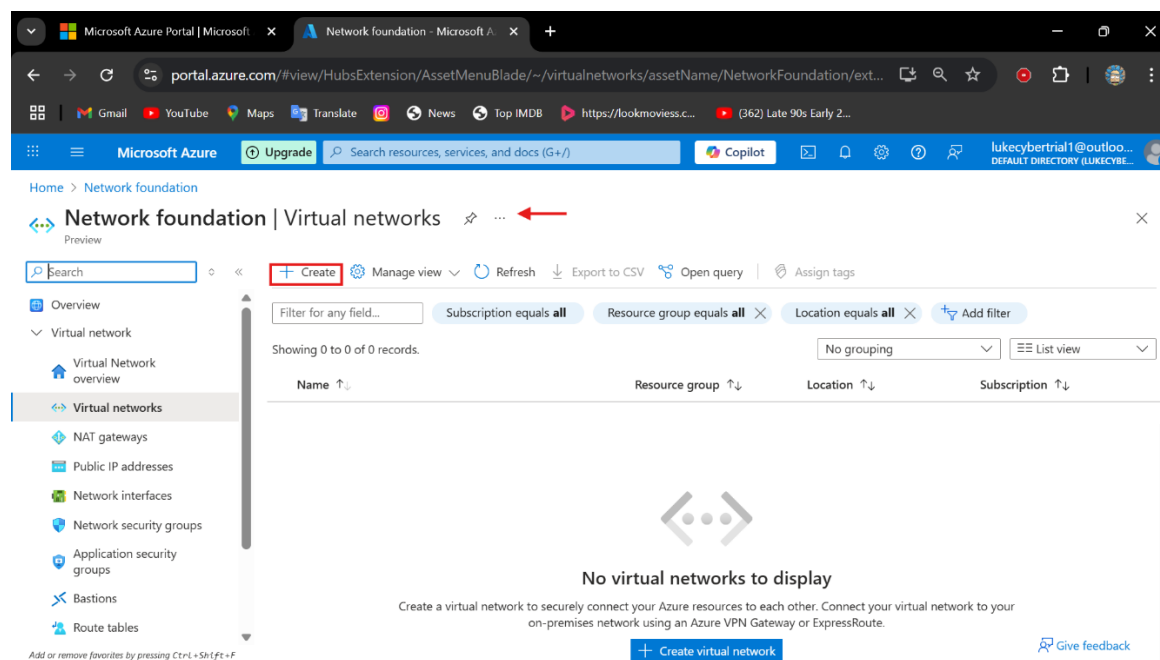
- **Web Servers** are accessible over HTTP from the internet but do **not allow RDP access**.
- **Management Servers** are accessible over **RDP only**, for administrative purposes.

Network Security Groups (NSGs) are used to enforce these access rules. The lab also includes testing to confirm that all traffic restrictions and access requirements function as expected.

Exercise 1: Create the virtual networking infrastructure

Task 1: Create a virtual network.

I started by signing in to the [Azure portal](#) using an account with Owner or Contributor rights. Once in, I used the search bar at the top to find **Virtual networks**. From there, I clicked **+ Create** to begin setting up a new virtual network, which I planned to use together with network and application security groups later in the lab.



After signing in to the Azure portal, I searched for **Virtual networks** and clicked on **+ Create** to set up a new one.

In the **Basics** tab:

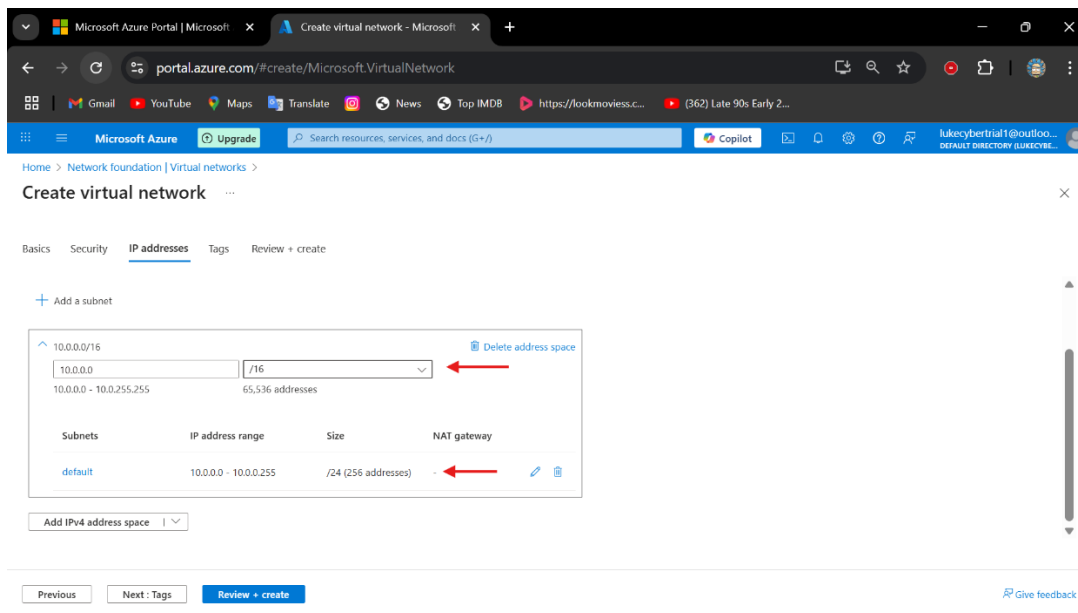
- Selected active Azure subscription1.
- Created a new resource group: AZ500LAB07.
- Named the VNet as myVirtualNetwork.
- Chose **East US** as the region.
- Clicked **Next: IP Addresses** to proceed.

The screenshot shows the 'Create virtual network' page in the Microsoft Azure Portal, specifically the 'Basics' tab. The page is titled 'Create virtual network' and has a breadcrumb trail: 'Home > Network foundation | Virtual networks >'. Below the title, there are tabs for 'Basics', 'Security', 'IP addresses', 'Tags', and 'Review + create'. The 'Basics' tab is active. Under 'Project details', there is a section 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.' It contains two dropdown menus: 'Subscription *' set to 'Azure subscription 1' and 'Resource group *' set to '(New) AZ500LAB07'. There is a 'Create new' link below the resource group dropdown. Under 'Instance details', there is a section for 'Virtual network name *' set to 'myVirtualNetwork' and 'Region *' set to '(US) East US'. There is a 'Deploy to an Azure Extended Zone' link below the region dropdown. At the bottom, there are buttons for 'Previous', 'Next: Security', and 'Review + create'. The 'Review + create' button is highlighted in blue. The bottom of the screen shows a Windows taskbar with various icons and a system tray showing '18°C' and '10:50 PM'.

In the **IP Addresses** tab:

- Set the IPv4 address space to 10.0.0.0/16.
- Clicked on the default subnet and edited it:
 - Subnet name: default
 - Subnet range: 10.0.0.0/24
 - Saved the subnet settings.

Went back to **Review + create**, reviewed the setup, and clicked **Create** to deploy the virtual network.

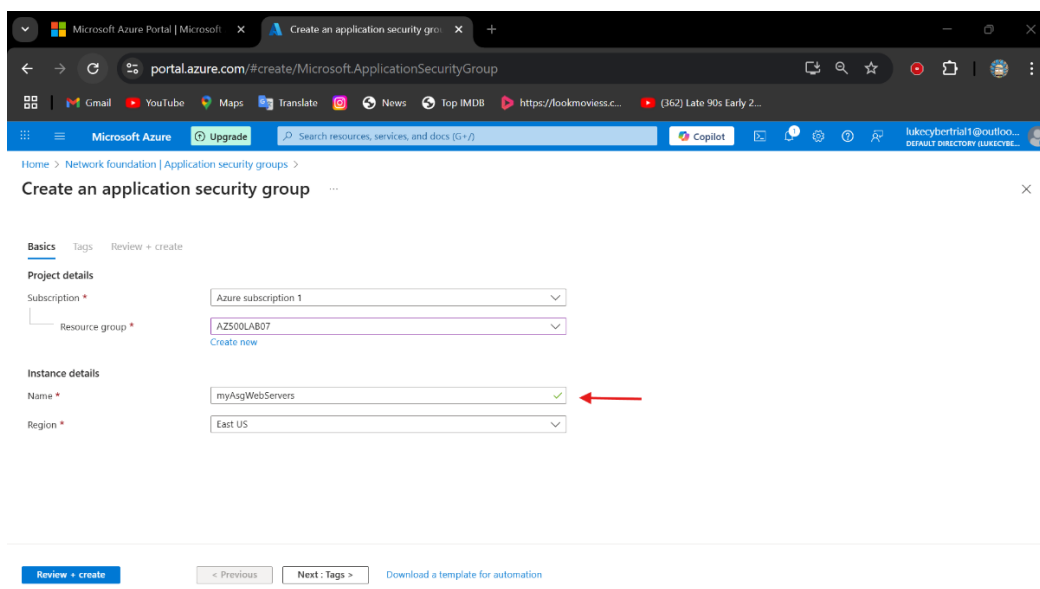


Task 2: Creating an Application Security Group

In the Azure portal, I searched for **Application security groups** and clicked on **+ Create** to set one up, on the **Basics** tab:

- Chose the existing resource group: **AZ500LAB07**
- Named the ASG: **myAsgWebServers**
- Selected region: **East US**
(*This group is intended for the web servers*)

Proceeded to **Review + create**, then clicked **Create** to deploy the security group.



Returned to the **Application security groups** blade and clicked **+ Create**. On the **Basics** tab:

- Resource group: AZ500LAB07
- Name: myAsgMgmtServers
- Region: **East US**
(This group is designated for the management servers)

Finalized by selecting **Review + create**, then clicked **Create**.

The screenshot shows the 'Create an application security group' blade in the Azure Portal. The 'Basics' tab is selected. Under 'Project details', 'Subscription' is set to 'Azure subscription 1' and 'Resource group' is set to 'AZ500LAB07'. Under 'Instance details', 'Name' is 'myAsgMgmtServers' (indicated by a red arrow) and 'Region' is 'East US'. At the bottom, there are buttons for 'Review + create', '< Previous', 'Next: Tags >', and 'Download a template for automation'. The system tray at the bottom shows a temperature of 18°C and the time 11:06 PM.

Task 3: Create a network security group and associate the NSG to the subnet

In this task, I created a **Network Security Group (NSG)** to control the inbound and outbound traffic within my Azure virtual network environment. NSGs help enforce security policies by allowing or denying traffic based on rules I define.

First, I went to the **Azure portal**, searched for **Network security groups**, and clicked on **+ Create** to start the process. On the **Basics** tab of the creation blade, I filled in the following details:

- **Subscription:** Azure subscription 1
- **Resource group:** AZ500LAB07

- **Name:** myNsg
- **Region:** East US

After confirming the inputs, I clicked **Review + create**, and finally **Create** to deploy the NSG. This NSG will later be associated with a subnet, allowing me to manage and restrict traffic at the subnet level based on the rules I define.

Microsoft Azure Portal | Microsoft | Create network security group - M

portal.azure.com/#create/Microsoft.NetworkSecurityGroup

Home > Network foundation | Network security groups >

Create network security group

Basics Tags Review + create

Project details

Subscription * Azure subscription 1

Resource group * AZ500LAB07 [Create new](#)

Instance details

Name * myNsg

Region * East US

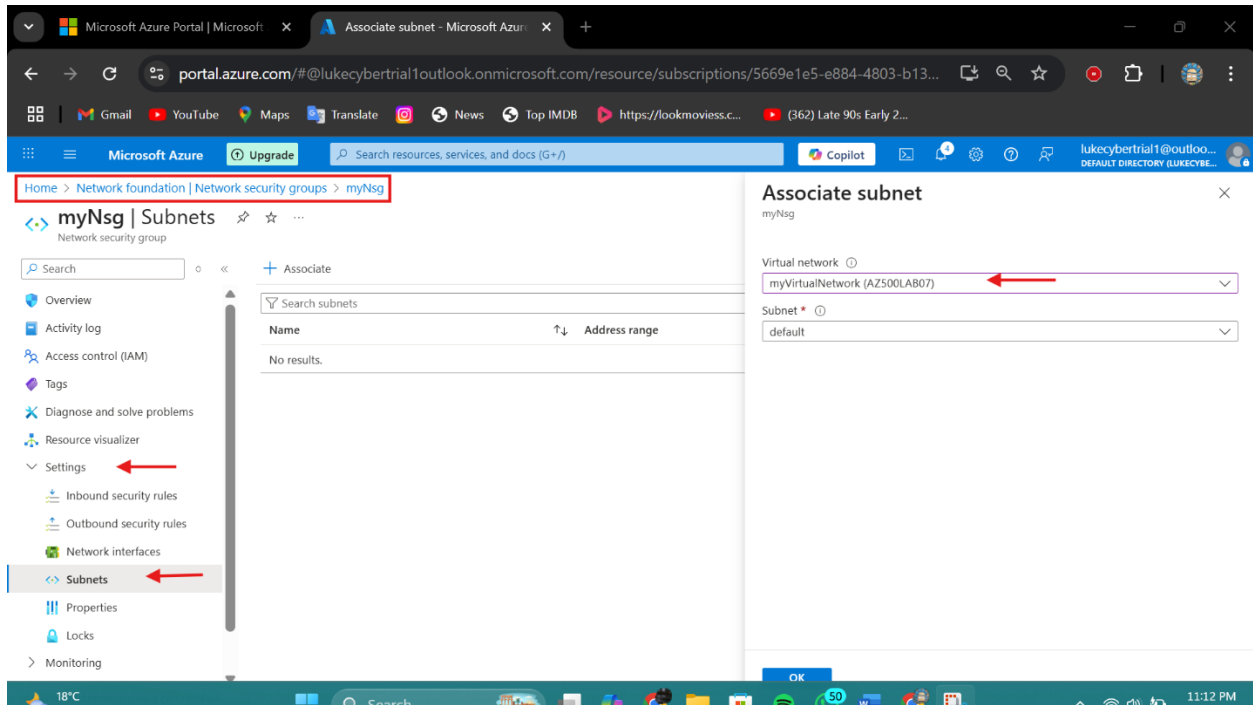
[Review + create](#) [< Previous](#) [Next: Tags >](#) [Download a template for automation](#)

To associate the Network Security Group (NSG) with a specific subnet, I followed these steps in the **Azure portal**. I navigated back to the **Network security groups** blade and selected the myNsg entry I had just created. On the myNsg blade, under the **Settings** section, I clicked on **Subnets** and then selected **+ Associate** to begin the association process.

On the **Associate subnet** blade, I provided the following settings:

- **Virtual network:** myVirtualNetwork
- **Subnet:** default

Once the settings were filled in correctly, I clicked **OK** to associate the NSG with the default subnet in the virtual network. This ensures that any traffic flowing through this subnet will now be subject to the security rules defined in myNsg.



Task 4: Create inbound NSG security rules.

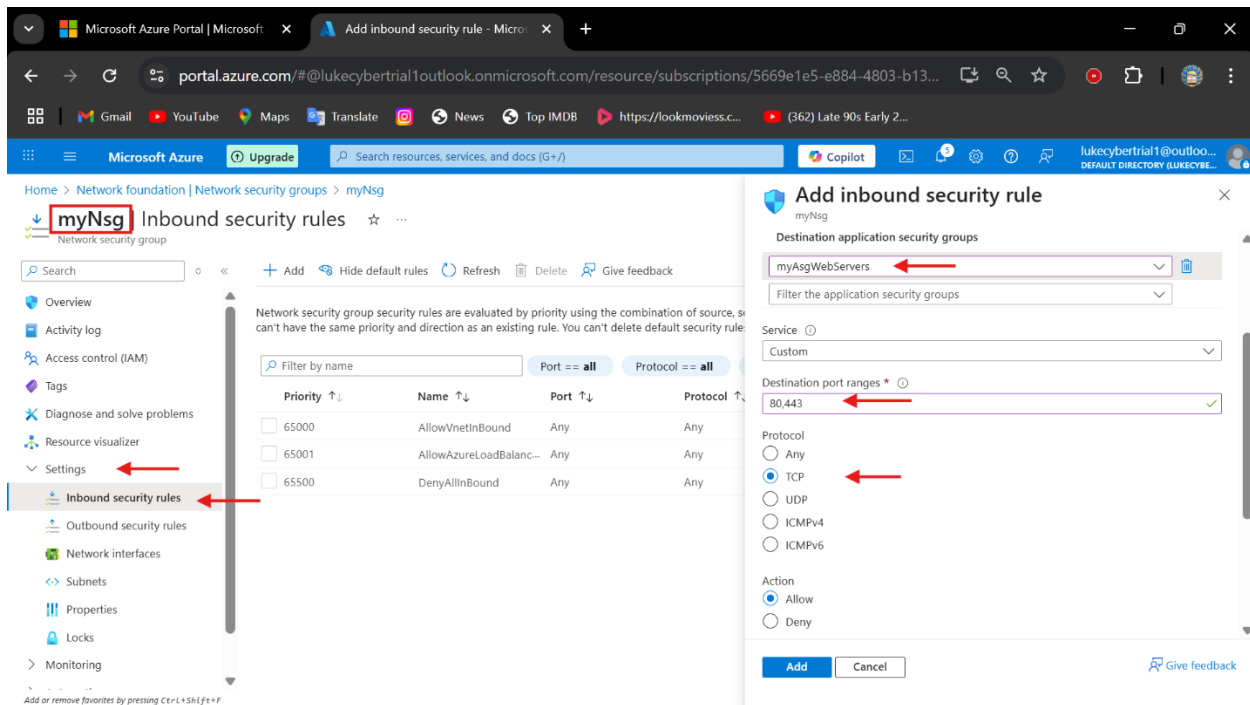
To allow web traffic to the web servers, I configured a new **inbound security rule** for the myNsg Network Security Group in the Azure portal.

From the myNsg blade, under the **Settings** section, I selected **Inbound security rules**. After reviewing the default rules, I clicked on **+ Add** to create a new rule.

On the **Add inbound security rule** blade, I used the following settings:

- **Destination:** Selected **Application security group**, then chose myAsgWebServers
- **Destination port ranges:** 80,443
- **Protocol:** TCP
- **Priority:** 100
- **Name:** Allow-Web-All

I left all other settings as default, then clicked **Add** to apply the rule. This rule now allows incoming TCP traffic on ports 80 (HTTP) and 443 (HTTPS) to all resources in the myAsgWebServers group.



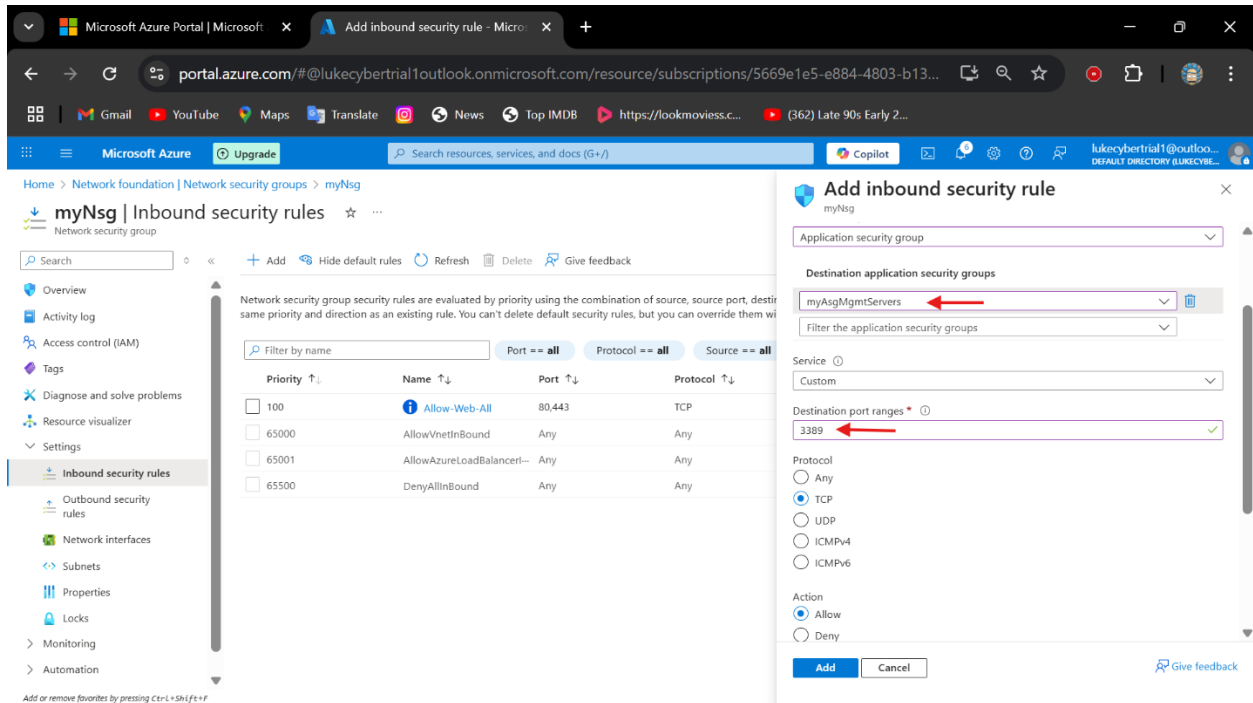
To enable remote access via RDP to the management servers, I added another **inbound security rule** to the myNsg Network Security Group.

From the **myNsg** blade, under the **Settings** section, I clicked **Inbound security rules**, then selected **+ Add**.

On the **Add inbound security rule** blade, I configured the rule with the following values:

- **Destination:** Chose **Application security group**, then selected myAsgMgmtServers
- **Destination port ranges:** 3389
- **Protocol:** TCP
- **Priority:** 110
- **Name:** Allow-RDP-All

I kept all other options at their default settings, then clicked **Add** to finalize the rule. This allows incoming RDP connections (port 3389) to all virtual machines within the myAsgMgmtServers application security group.



Exercise 2: Deploy virtual machines and test network filters

Task 1: Create a virtual machine to use as a web server.

To create the **myVmWeb** virtual machine that will act as a web server, I opened the Azure portal and searched for **Virtual machines** using the top search bar, then pressed **Enter**. On the **Virtual machines** blade, I clicked **+ Create** and selected **+ Azure virtual machine** from the dropdown. On the **Basics** tab of the **Create a virtual machine** blade, I configured the following settings:

- **Subscription:** the name of the Azure subscription used in this lab
- **Resource group:** AZ500LAB07
- **Virtual machine name:** myVmWeb
- **Region:** (US) East US
- **Availability options:** No infrastructure redundancy required
- **Security type:** Standard
- **Image:** Windows Server 2022 Datacenter: Azure Edition – x64 Gen2
- **Size:** Standard D2s v3

- **Username:** Student
- **Password:** Student2025!
- **Confirm password:** Student2025!
- **Public inbound ports:** None
- **Would you like to use an existing Windows Server License:** No

Microsoft Azure Portal | Microsoft x Create a virtual machine - Microso x +

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure Upgrade Search resources, services, and docs (G+7) Copilot lukecybertial1@outloo... DEFAULT DIRECTORY BURECTM...

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Virtual machine name * myVmWeb

Region * (US) East US

Availability options No infrastructure redundancy required

Security type Standard

Image * Windows Server 2022 Datacenter: Azure Edition - x64 Gen2 (free services el)

See all images | Configure VM generation

This image is compatible with additional security features. Click here to swap to the Trusted launch security type.

VM architecture

Arm64

x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount

< Previous Next: Disks > Review + create Give feedback

Microsoft Azure Portal | Microsoft x Create a virtual machine - Microso x +

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure Upgrade Search resources, services, and docs (G+7) Copilot lukecybertial1@outloo... DEFAULT DIRECTORY BURECTM...

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Size * Standard_DS2s_v3 - 2 vcpus, 8 GiB memory (\$137.24/month)

See all sizes

Enable Hibernation

Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. Learn more

Administrator account

Username * Student

Password *

Confirm password *

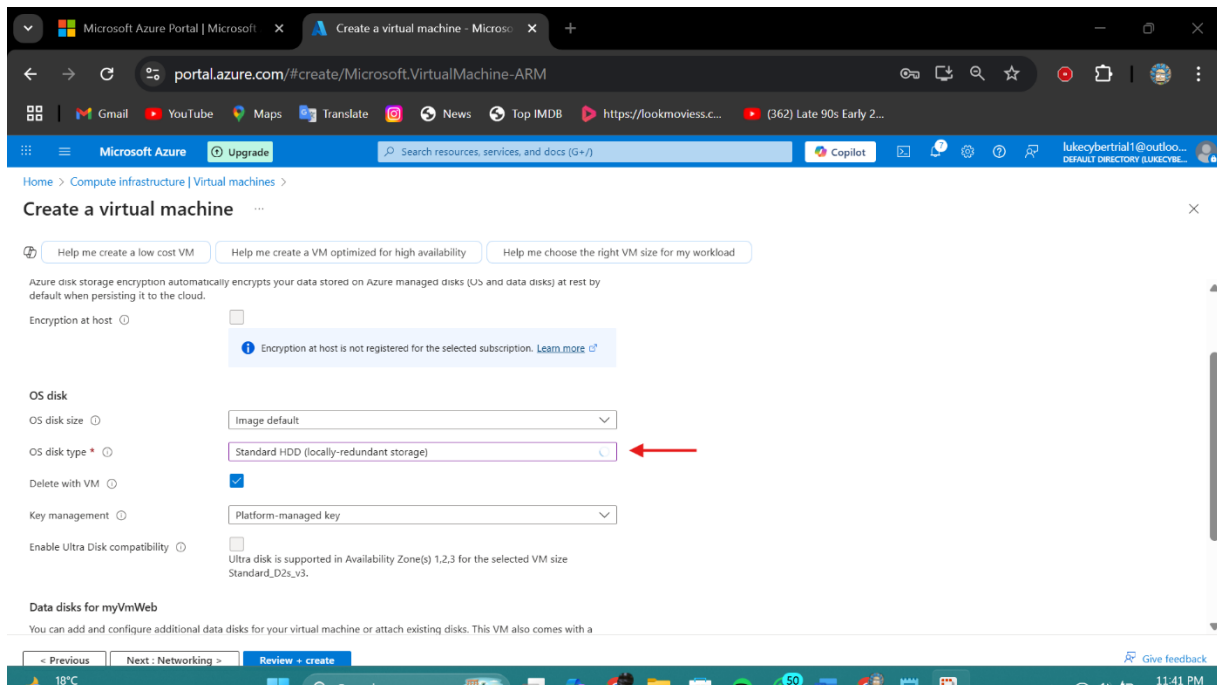
Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

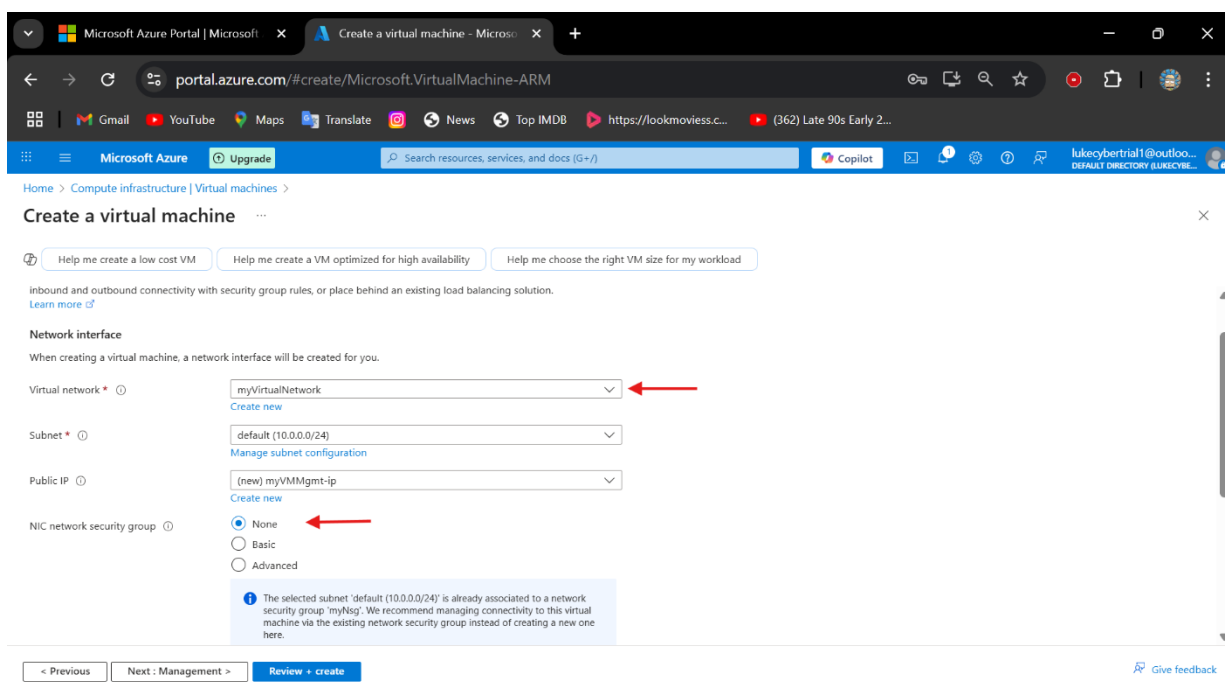
Public inbound ports * None

< Previous Next: Disks > Review + create Give feedback

On the **Disks** tab of the **Create a virtual machine** blade, I clicked **Next: Disks >**, then set the **OS disk type** to **Standard HDD**. After making this change, I clicked **Next: Networking >** to proceed to the next configuration step.



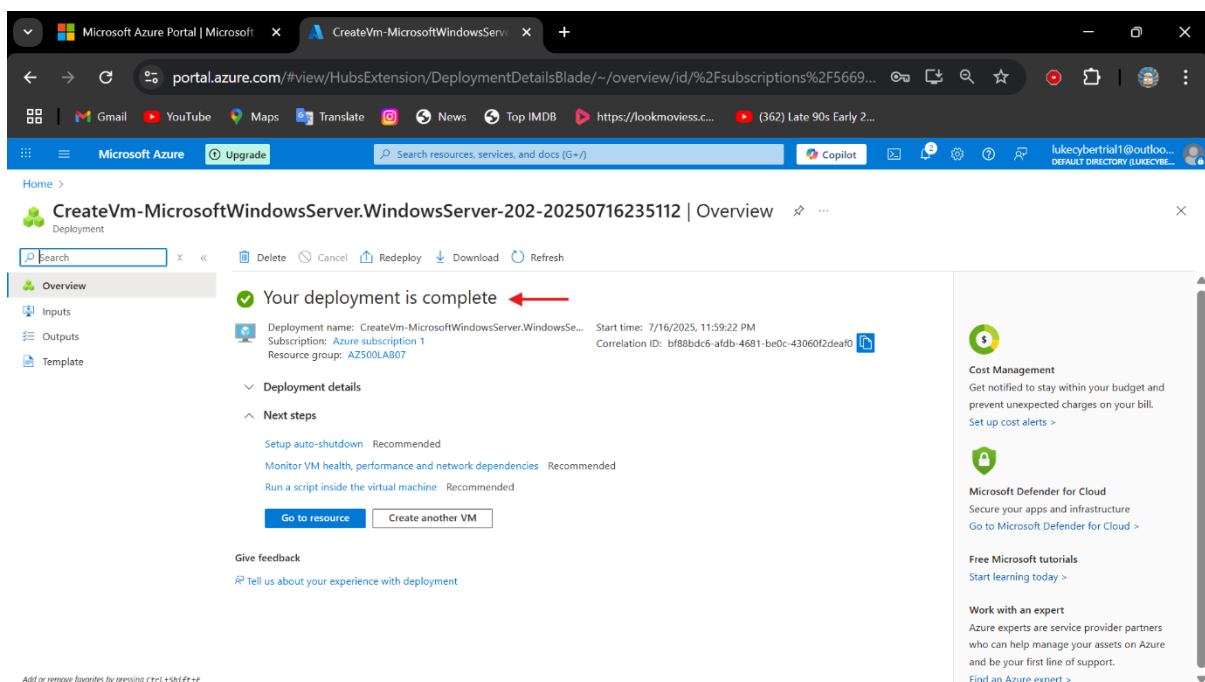
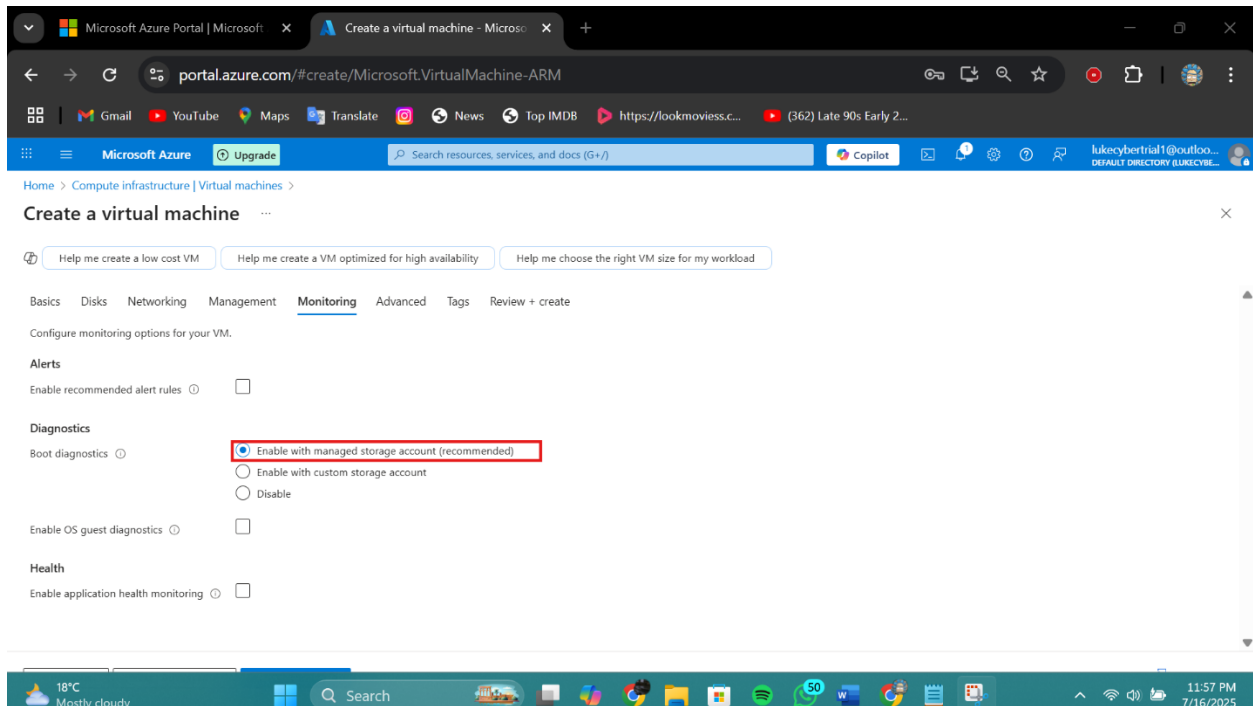
On the **Networking** tab of the **Create a virtual machine** blade, I selected the previously created virtual network named **myVirtualNetwork**. Under **NIC network security group**, I set the value to **None**.



Next, I clicked **Next: Management >**, then **Next: Monitoring >**. On the **Monitoring** tab of the *Create a virtual machine* blade, I verified that the following setting was applied:

- **Boot diagnostics:** Enabled with managed storage account (recommended)

After confirming the setting, I clicked **Review + create**, ensured that validation was successful on the *Review + create* blade, and then clicked **Create** to deploy the virtual machine.



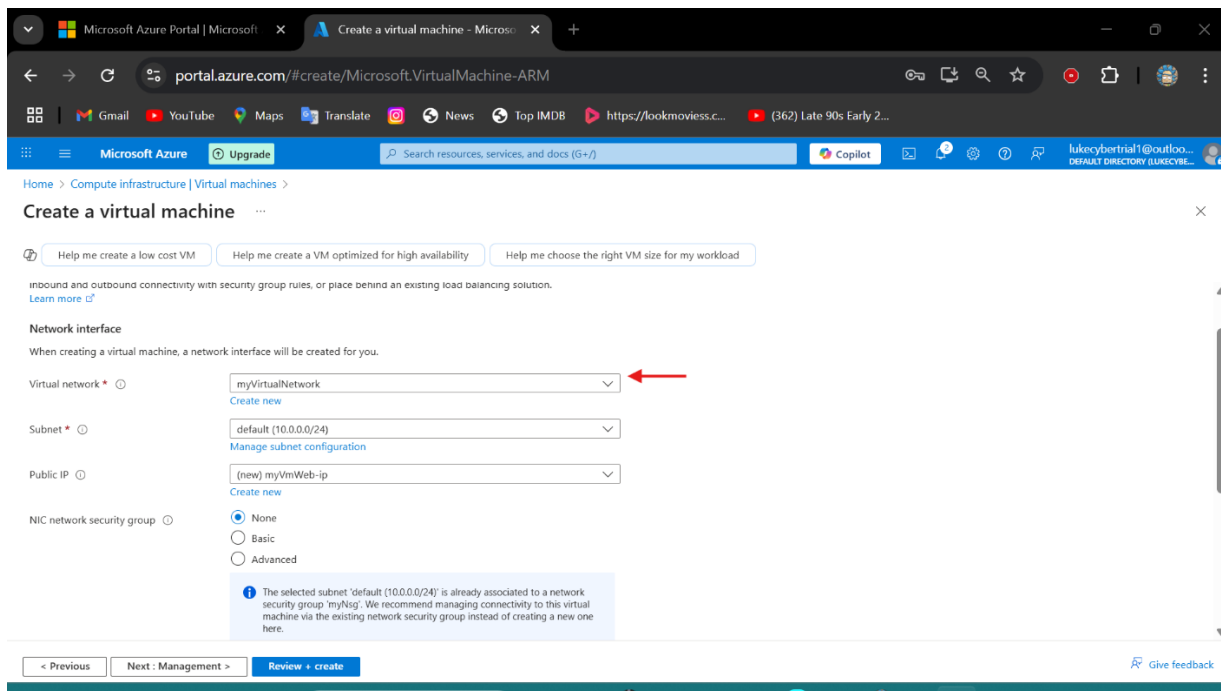
Task 2: Create a virtual machine to use as a management server.

To create the **myVMMgmt** virtual machine that will act as a management server. In the Azure portal, I navigated back to the **Virtual machines** blade, clicked **+ Create**, and selected **+ Azure virtual machine** from the dropdown list. On the **Basics** tab of the *Create a virtual machine* blade, I configured the following settings:

- **Subscription:** The name of the Azure subscription I am using in this lab
- **Resource group:** AZ500LAB07
- **Virtual machine name:** myVMMgmt
- **Region:** (US) East US
- **Availability options:** No infrastructure redundancy required
- **Security type:** Standard
- **Image:** Windows Server 2022 Datacenter: Azure Edition - x64 Gen2
- **Size:** Standard D2s v3
- **Username:** Student
- **Password:** My personal password from Lab 02 > Exercise 2 > Task 1 > Step 3
- **Public inbound ports:** None
- **Already have a Windows Server license:** No

I then proceeded to click **Next: Disks >**, and on the **Disks** tab of the *Create a virtual machine* blade, set the **OS disk type** to **Standard HDD**, then click **Next: Networking >** to proceed.

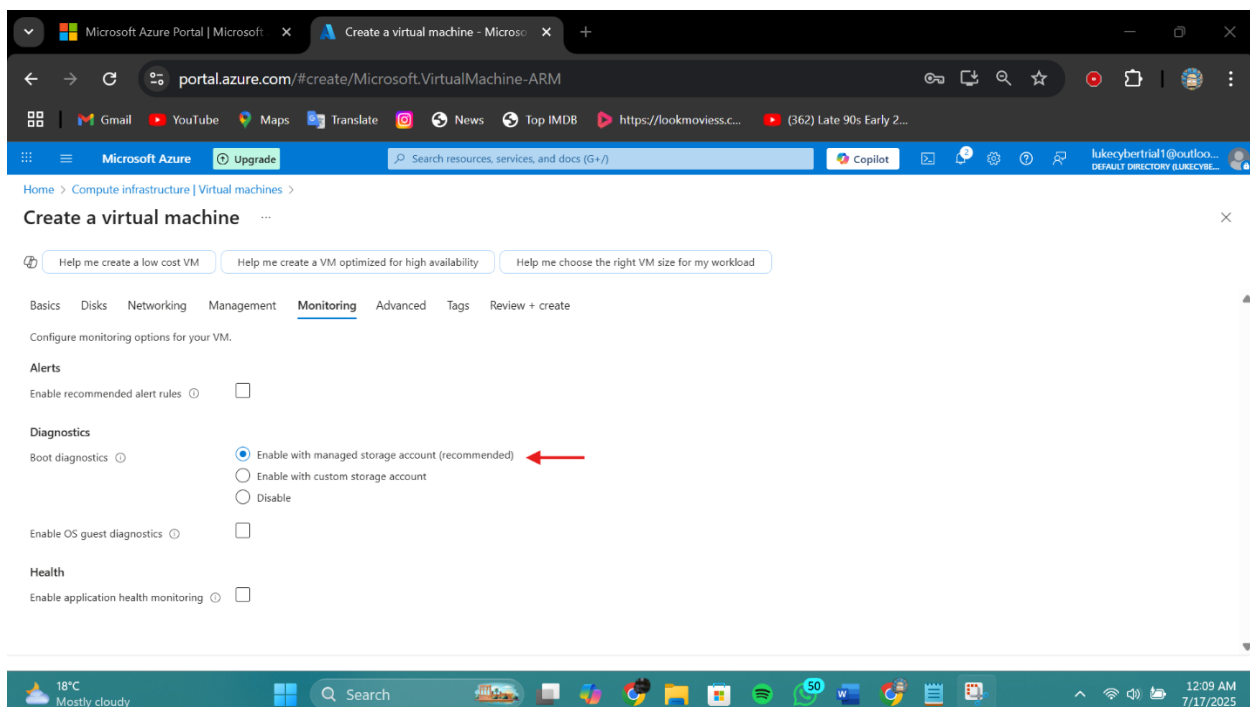
On the **Networking** tab of the *Create a virtual machine* blade, select the previously created virtual network **myVirtualNetwork**. Under **NIC network security group**, choose **None** to proceed without assigning a network security group to the network interface.



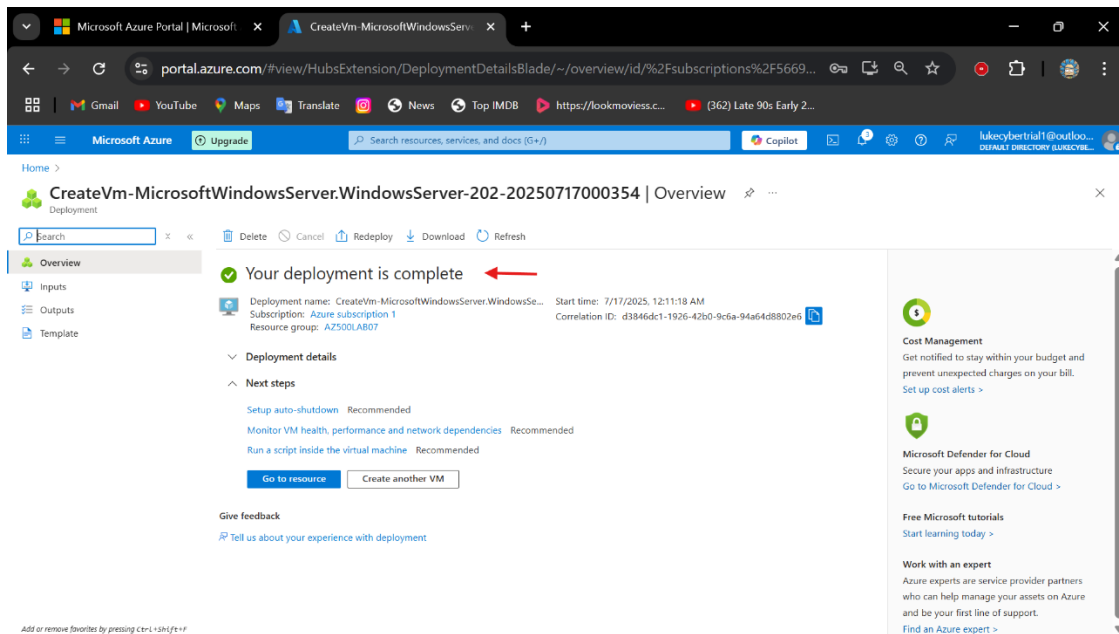
I clicked **Next: Management >**, then **Next: Monitoring >**.

On the **Monitoring** tab of the *Create a virtual machine* blade, I verified the following setting:

- **Boot diagnostics:** *Enabled with managed storage account (recommended)*



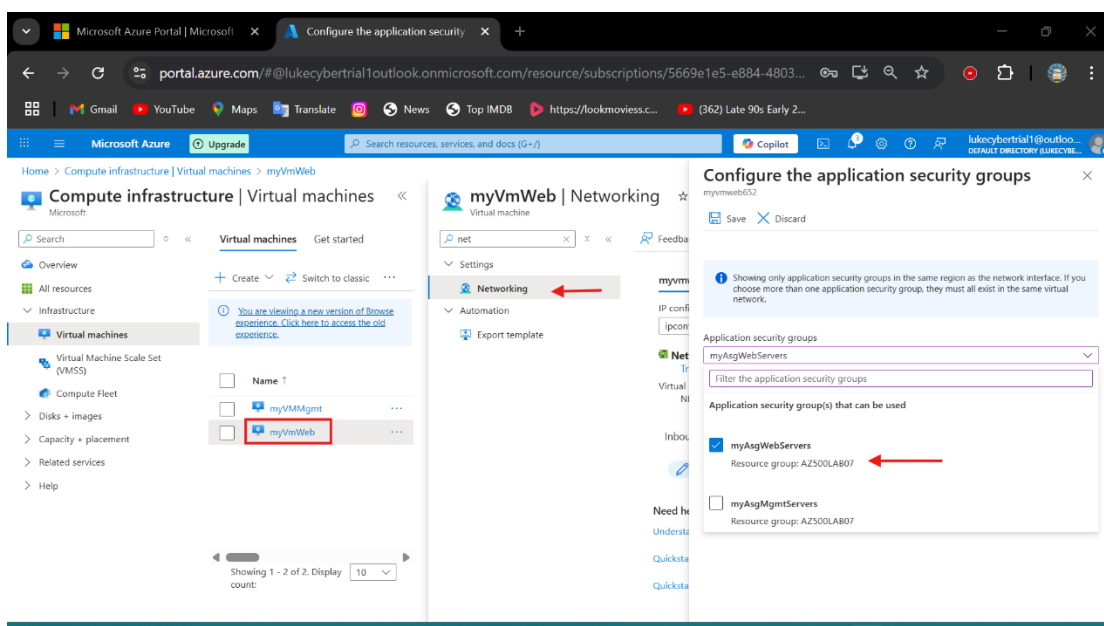
I then clicked **Review + create**, ensured that validation was successful, and clicked **Create** to deploy the virtual machine.



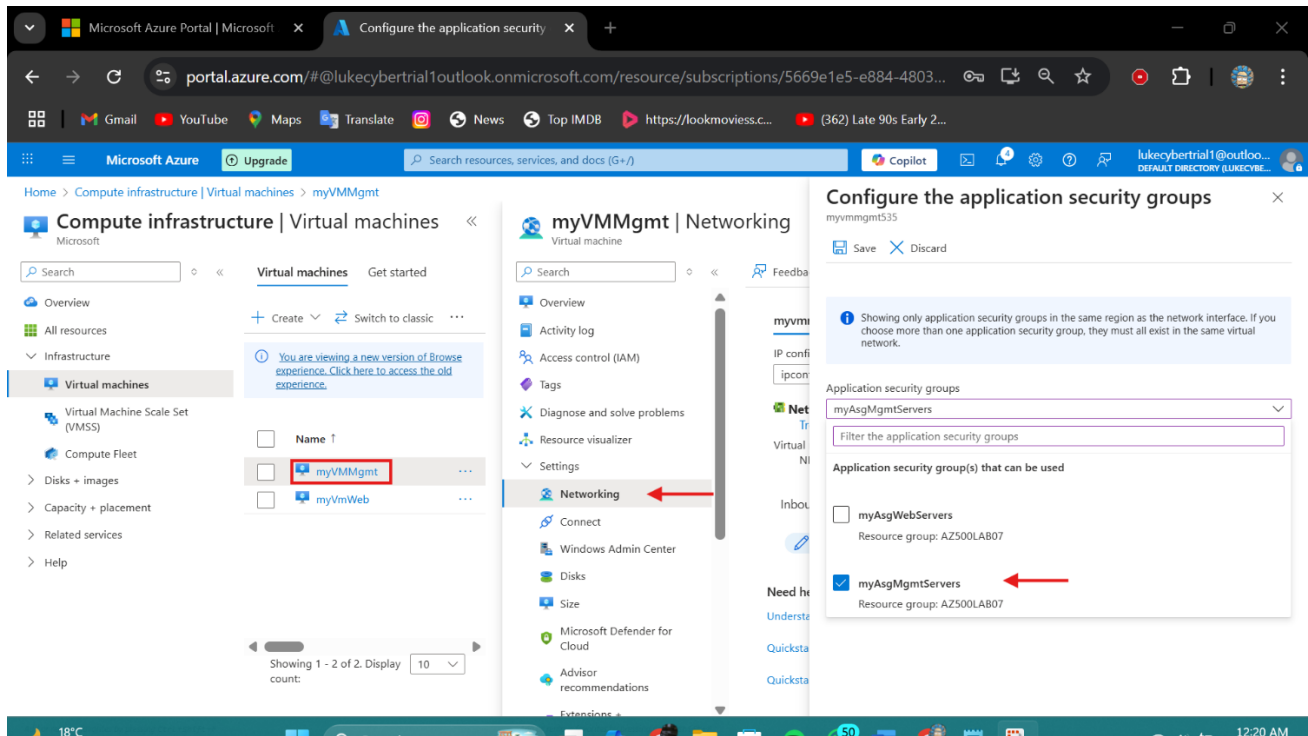
Task 3: Associate each virtual machines network interface to its application security group.

I navigated to the Virtual machines blade in the Azure portal and verified that both virtual machines—myVMWeb and myVMMgmt—were listed with the **Running** status.

I clicked on the **myVMWeb** entry, went to the **Networking** section, and selected **Network settings**. On the **myVMWeb | Networking settings** blade, I switched to the **Application security groups** tab. I clicked **+ Add application security groups**, selected **myAsgWebServers** from the list, and clicked **Save** to associate the network interface with the ASG.



I navigated back to the **Virtual machines** blade and clicked on the **myVMMgmt** entry from the list. On the **myVMMgmt** blade, I went to the **Networking** section and selected **Network settings**. In the **myVMMgmt | Networking settings** blade, I opened the **Application security groups** tab. I clicked **+ Add application security groups**, selected **myAsgMgmtServers** from the list, and clicked **Save** to complete the association.



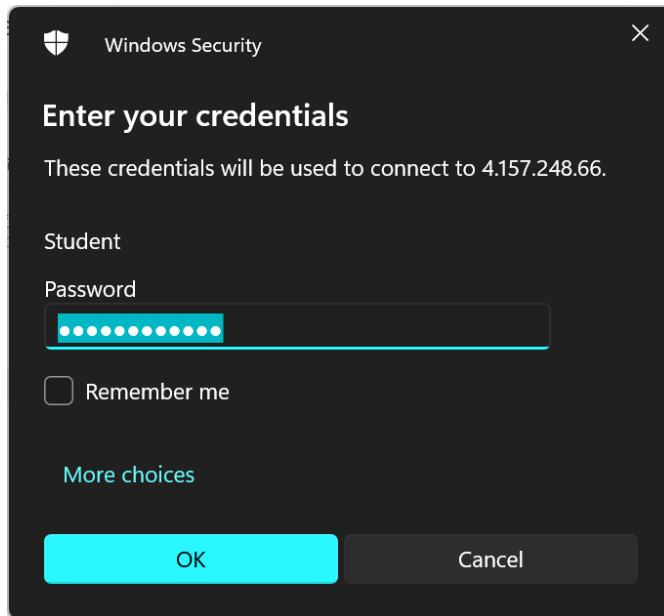
Task 4: Test the network traffic filtering

I navigated back to the **myVMMgmt** virtual machine blade. On the **Overview** tab, I clicked **Connect**, then selected **Connect** again from the dropdown menu.

I downloaded the RDP file and used it to initiate a Remote Desktop connection to the **myVMMgmt** Azure VM. When prompted, I entered the following credentials:

- **User name:** Student
- **Password:** *Student2025!*

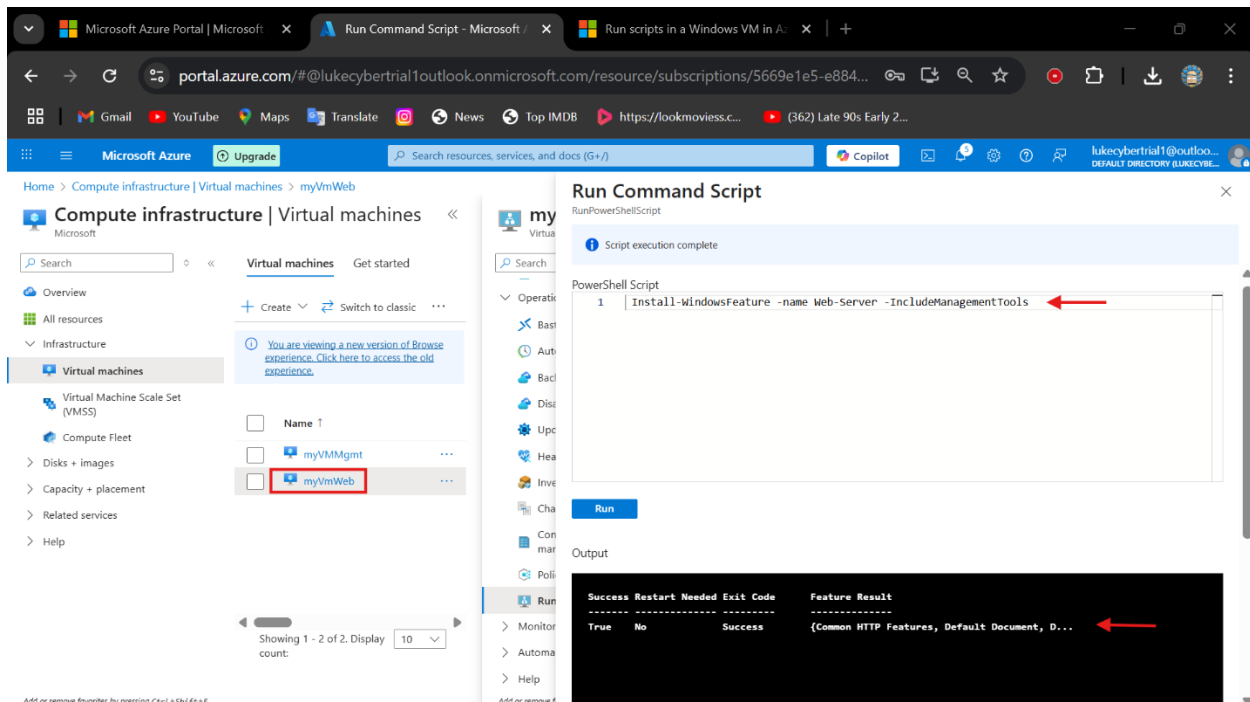
I was successfully able to connect to the **myVMMgmt** VM via RDP.



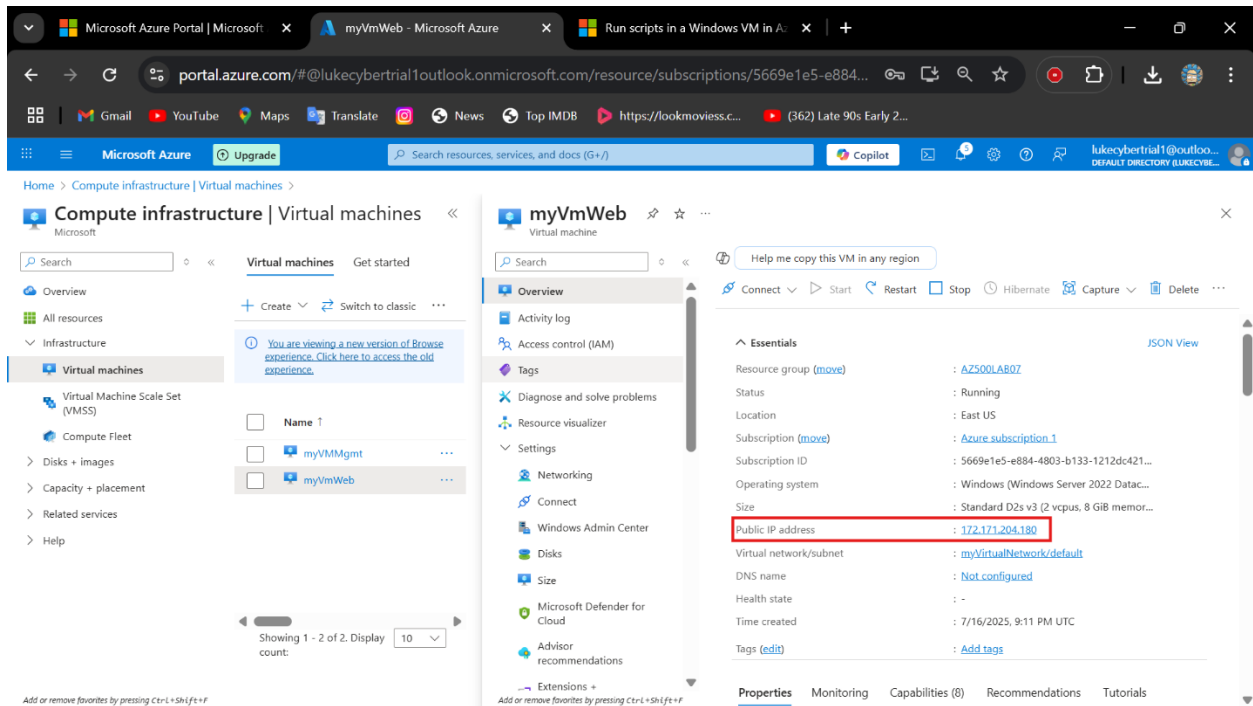
I navigated to the **myVMWeb** virtual machine blade in the Azure portal. Under the **Operations** section, I selected **Run command**, then clicked **RunPowerShellScript**. In the script pane, I ran the following command to install the Web Server (IIS) role:

[Install-WindowsFeature -name Web-Server -IncludeManagementTools]

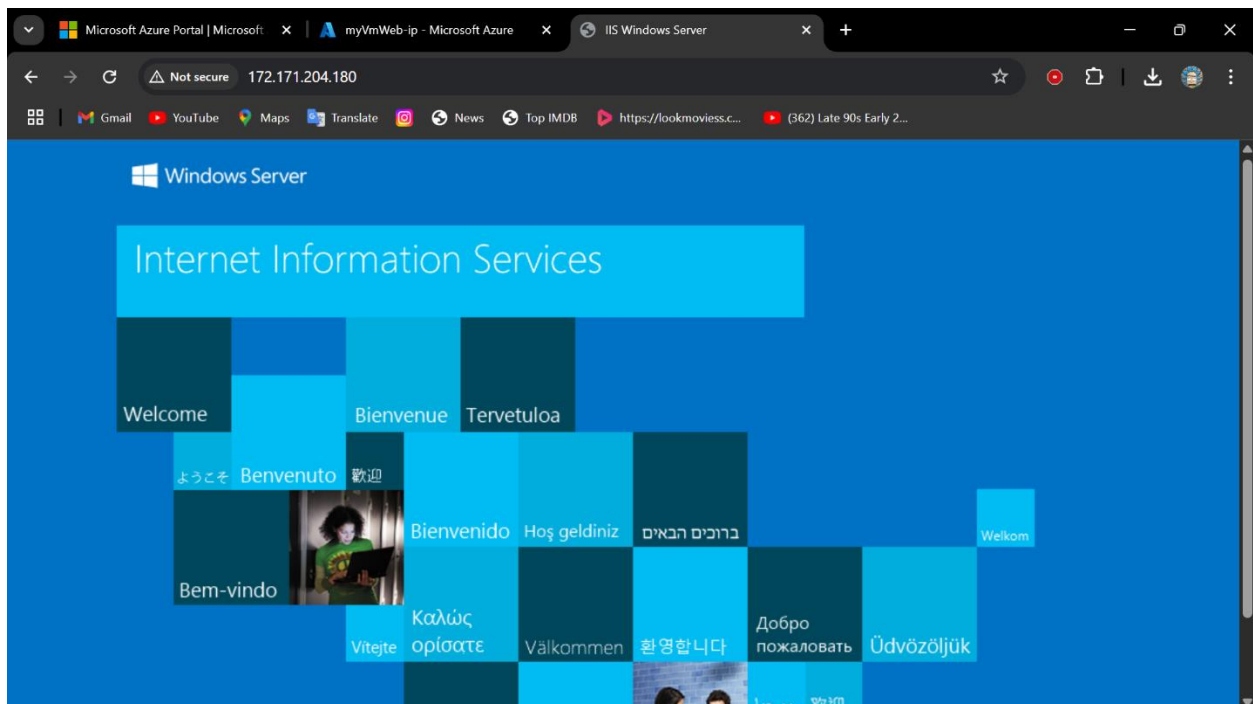
I waited for the installation to complete. Once done, I verified that **myVMWeb** could be accessed via **HTTP/HTTPS**, confirming that the IIS web server was installed and running.



I navigated back to the **myVmWeb** blade in the Azure portal and located the **Public IP address** of the virtual machine.



I then opened a new browser tab and entered the IP address in the address bar. The browser successfully loaded the **default IIS web page**, confirming that the web server is accessible from the internet.



Clean-up.

I opened the **Cloud Shell** in the Azure portal and selected **PowerShell**. When prompted, I created the storage if it wasn't already set up. In the PowerShell session, I ran the following command to clean up all resources created during the lab: [Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob]. This initiated the deletion of the **AZ500LAB07** resource group and all associated resources, helping avoid any unnecessary charges.

The screenshot shows a PowerShell session in the Azure Cloud Shell. The user has executed the command `Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob`. The session displays three tables showing the progress of the job. The first two tables show the job in a 'Running' state, and the third table shows it as 'Completed'.

```
PS /home/luke> Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob
```

ID	Name	PSJobTypeName	State	HasMoreData	Location	Command
1	Long Running O...	AzureLongRunni...	Running	True	localhost	Remove-AzResourceGroup

```
PS /home/luke> Get-Job
```

ID	Name	PSJobTypeName	State	HasMoreData	Location	Command
1	Long Running O...	AzureLongRunni...	Running	True	localhost	Remove-AzResourceGroup

```
PS /home/luke> Get-Job
```

ID	Name	PSJobTypeName	State	HasMoreData	Location	Command
1	Long Running O...	AzureLongRunni...	Completed	True	localhost	Remove-AzResourceGroup

```
PS /home/luke> Remove-AzResourceGroup -Name "NetworkWatcherRG" -Force -AsJob
```

ID	Name	PSJobTypeName	State	HasMoreData	Location	Command
2	Long Running O...	AzureLongRunni...	Running	True	localhost	Remove-AzResourceGroup

```
PS /home/luke> Get-Job
```

ID	Name	PSJobTypeName	State	HasMoreData	Location	Command
1	Long Running O...	AzureLongRunni...	Completed	True	localhost	Remove-AzResourceGroup
2	Long Running O...	AzureLongRunni...	Completed	True	localhost	Remove-AzResourceGroup

Lab Summary

1. Created Two Virtual Machines:

- **myVMWeb** and **myVMMgmt** were deployed in the AZ500LAB07 resource group within the same virtual network (myVirtualNetwork) and subnet.

2. Configured Networking:

- Set **NIC network security group** to **None** for both VMs during creation.
- Verified that both VMs were in **Running** state.

3. Created Application Security Groups (ASGs):

- Created **myAsgWebServers** and **myAsgMgmtServers** to logically group the VM NICs based on function.

4. Associated NICs with ASGs:

- Linked **myVMWeb**'s NIC to **myAsgWebServers**.
- Linked **myVMMgmt**'s NIC to **myAsgMgmtServers** via the Networking settings blade.

5. Tested Traffic Filters:

- **RDP** connection was tested and successful to **myVMMgmt**.
- Installed IIS on **myVMWeb** using PowerShell: *[Install-WindowsFeature -name Web-Server -IncludeManagementTools]*
- Accessed the **default IIS web page** via **myVMWeb's public IP address** from a browser — verified successful HTTP traffic.

6. Cleaned Up Resources:

- Deleted the entire lab setup using: *[Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob]*

Conclusion

This lab demonstrated how to enhance network security in Azure using **Application Security Groups (ASGs)**. By logically grouping virtual machine NICs and applying network security rules based on ASG membership, we simplified traffic management between tiers of a multi-VM deployment. We successfully verified connectivity using **RDP** and **HTTP**, then confirmed our configuration by accessing the IIS web server. Finally, we cleaned up resources to avoid unnecessary costs. This exercise reinforced the value of ASGs in implementing scalable and maintainable network security in the cloud.